BLOCK 6

Non-Overlapping Clock Generator

From a fixed-bias baseline to a PVT-tracking bias generator, with mismatch Monte Carlo

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Technology	SKY130A, 1.8 V MOS devices, ngspice 46
Clock	4 MHz, $T = 250$ ns
Circuit	80 MOS NONOV + 3 MOS and 1 resistor bias generator
Evidence	45 PVT conditions; 3 x 100 Monte Carlo seeds; 12 start-up runs
Revision	1.0 / 9 November 2026

Result status. The block meets the screening criteria in all 45 PVT conditions and in the nominal closed loop. With local mismatch, 70 of 100 samples meet the width criterion at the worst corner. This is a pre-layout simulation result, not a sign-off, not a post-layout result, and the criteria are my own screening limits.

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Summary for the reader

Block 6 turns one 4 MHz clock into the two non-overlapping phases ϕ_1 and ϕ_2 (and their complements) that drive the switches of the modulator. The topology follows Spilka et al. [1]: XOR, current-starved delay cells, AND2, XOR, with no feedback loop. The netlist has 80 MOS devices. The delay cells set the *gap* between the phases, and the gap is set by their bias current.

The first version used ideal fixed bias voltages ($V_{BN} = 0.70$ V, $V_{BP} = 0.71$ V). At the nominal point it gave a gap of 14.3 ns and a phase width of 110.6 ns with no overlap. Over the 45 PVT conditions (five corners, $-40/27/125\,^{\circ}$C, 1.62/1.8/1.98 V), however, only **17** conditions passed the screening criteria. Ten conditions produced no phases at all, and the gap ranged from 1.94 ns to 91.76 ns.

A bias generator that tracks PVT fixes this. It uses one resistor, an NMOS diode, an NMOS mirror and a PMOS replica diode, and sets the delay-cell current as $I = (V_{DD} - V_{BN})/R$. With it, **45 of 45** conditions pass, the gap lies between 9.99 ns and 20.58 ns, and the smallest phase width is 104.3 ns. Start-up from a supply ramp settles in two to four pulses in 12 of 12 runs, and the comparator clock taken from ϕ_{2b} rises inside the gap in 45 of 45 conditions. In the closed loop at the nominal point the generator gives SNDR 73.62 dB at OSR 80, which is within the single-run scatter of the other configurations.

Mismatch changes the picture. A local-mismatch Monte Carlo of 100 seeds per point gives 99/100 passing samples at TT, 100/100 at the smallest-gap corner, and **70/100** at the smallest-width corner (FS, $-40\,^{\circ}$C, 1.62 V). All 30 failures there have a phase width below the 100 ns screening limit, with a worst case of 67.1 ns. This limit is not derived from the settling budget of Block 5, so the result is a known limit and not a proven failure of the system.

Decision. Close Block 6 with the known limits listed in Section 5. The next iteration, if time allows, is to re-centre the resistor, enlarge the generator mirror devices, and derive the width requirement from Block 5 settling. Any change of R invalidates every result in this report.

1 Purpose and scope

1.1 Role of Block 6 in the modulator

The modulator is a second-order switched-capacitor delta-sigma loop with a 1.8 V supply, a 4 MHz sampling rate, and OSR 80. The clock period is $T = 250\,\text{ns}$, so each phase has at most $T/2 = 125\,\text{ns}$. Block 5 (switch network) uses ϕ_1 and ϕ_2 to sample and transfer charge; the StrongARM comparator of Block 3 needs a clock that arrives while neither phase is active. A switched-capacitor integrator uses both clock phases, so the shorter of the two limits the time available to each half-cycle [1]; the non-overlap gap prevents charge from leaking between the two configurations.

1.2 Questions this report answers

- Does the generator produce clean, non-overlapping phases at the nominal point?
- Does it keep working over process, voltage and temperature, and why does a fixed bias not?
- Does a bias generator that tracks PVT repair the corner conditions, and at what cost in spread?
- How much does local mismatch change gap and width?
- Does the block start from a supply ramp, and does the derived comparator clock fall inside the gap?
- Does the closed loop still work with these phases?

1.3 What the report does not claim

All results come from pre-layout ngspice simulation. Status labels are `MEASURED_NOT_FINAL` and `SIMULATED_NOT_FINAL_PASS`. The acceptance criteria in Table 2 are screening thresholds that I chose; they are not derived from a switch-settling or comparator-timing budget. A single closed-loop SNDR value has a scatter of about ±1 dB at OSR 80, so differences of that size are not claimed as degradation or improvement.

2 Circuit and theory

2.1 Non-overlap principle

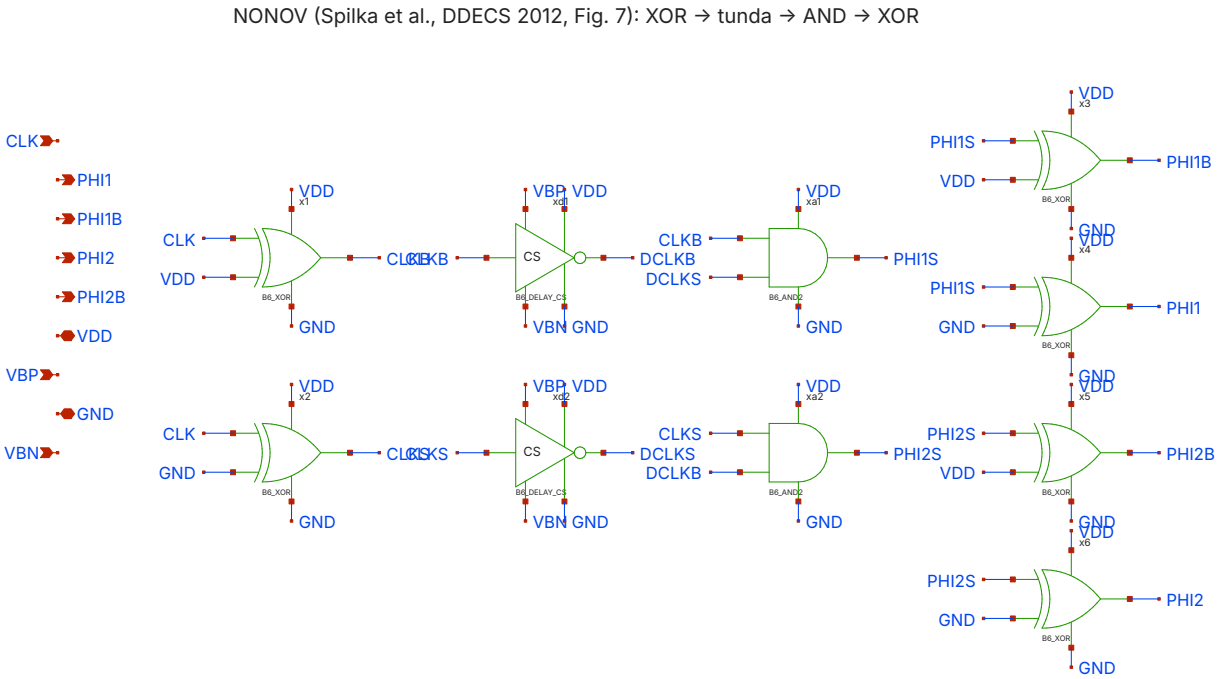


Figure 1: Xschem export of `B6_NONOV.sch`. The first XOR stages make CLKB (inverted clock) and CLKS (buffered clock), the delay cells `B6_DELAY_CS` make DCLKB and DCLKS, the AND2 cells make ϕ_1S and ϕ_2S, and the last XOR stages produce the phases and their complements. The title text is part of the schematic file.

In the netlist, the first XOR stages make CLKB (inverted clock) and CLKS (buffered clock). Each delay cell is an inverter, so DCLKB and DCLKS are delayed inverted copies. The two AND2 gates then form $\phi_1\mathrm{S} = \mathrm{CLKB} \cdot \mathrm{DCLKS}$ and $\phi_2\mathrm{S} = \mathrm{CLKS} \cdot \mathrm{DCLKB}$. Because the delayed copy lags the clock, each pulse rises one delay time after the corresponding clock edge and falls with the clock. The last XOR stages produce the phase and its complement.

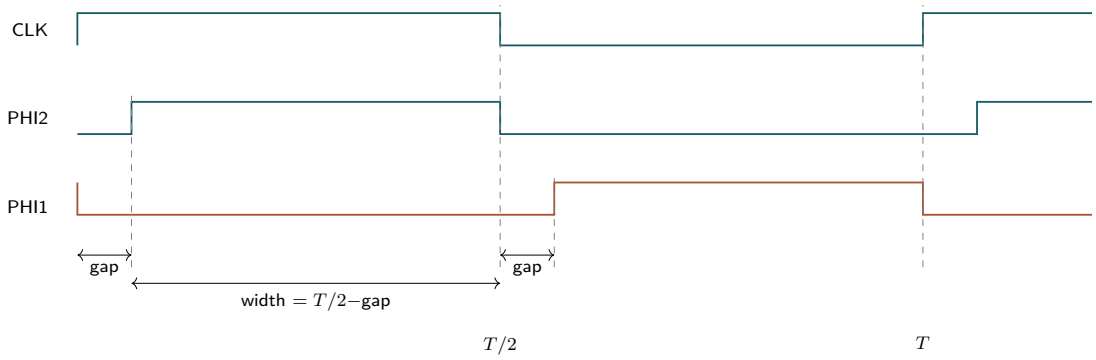


Figure 2: Idealised timing (not to scale; the gap is exaggerated). Each phase rises one gap after its clock edge and falls with it.

This gives, to first order,

$$t_{gap} \approx t_d, \qquad t_{width} \approx \frac{T}{2} - t_d, \qquad t_{gap} + t_{width} \approx \frac{T}{2} = 125\,\mathrm{ns}. \tag{1}$$

The sum in (1) holds in the data: in every deterministic PVT run, gap plus width is between 124.7 ns and 125.0 ns. The consequence is a direct trade-off. A larger gap protects against overlap but takes width from each phase, and the screening limits in Table 2 bracket the gap from both sides.

2.2 Delay cell and bias sensitivity

Each delay cell is a current-starved inverter: an NMOS starving device ($L = 1\,\mu$m, $W = 0.5\,\mu$m, gate V_{BN}) and a PMOS starving device ($L = 1\,\mu$m, $W = 1\,\mu$m, gate V_{BP}), a minimum-length inverter, and a MIM capacitor at the output. To first order, a delay is $t_d \approx C\,\Delta V / I$, so t_{gap} scales roughly as $1/I$. The simulations of a single cell show how steep this is:

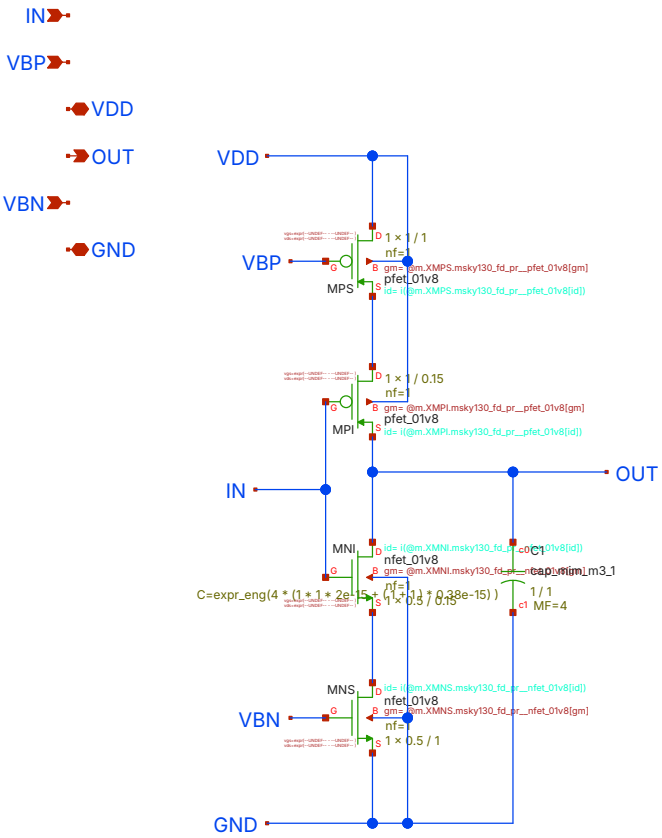


Figure 3: Xschem export of `B6_DELAY_CS.sch`: current-starved inverter with PMOS MPS (gate VBP), PMOS MPI, NMOS MNI, NMOS MNS (gate VBN), and the MIM capacitor C1 on OUT. The device annotations are the Xschem probe labels.

Table 1: Single delay cell versus V_{BN}, TT, 27 °C, 1.8 V (nominal characterisation).

V_{BN} (V)	0.60	0.70	0.80
Rising / falling delay (ns)	116.1 / 104.3	22.1 / 20.9	7.4 / 7.3

A change of 100 mV in the gate bias changes the delay by a factor of about 5. A bias that does not follow supply and temperature therefore cannot hold the gap.

2.3 Why a fixed bias fails over PVT

With V_{BN} and V_{BP} fixed, the current of the starving devices depends on the threshold voltage and on V_{DD}. The data show the expected direction: cold and low-supply conditions slow the cell (gap up to 91.76 ns at SS, −40 °C, 1.8 V), and hot, high-supply conditions speed it up (gap down to 1.94 ns at FS, −40 °C, 1.98 V). This is consistent with a threshold shift at a fixed gate voltage; I did not test the explanation separately. At 1.62 V, nine of the ten conditions without phases occur, so the fixed bias is unusable at low supply.

2.4 Bias generator

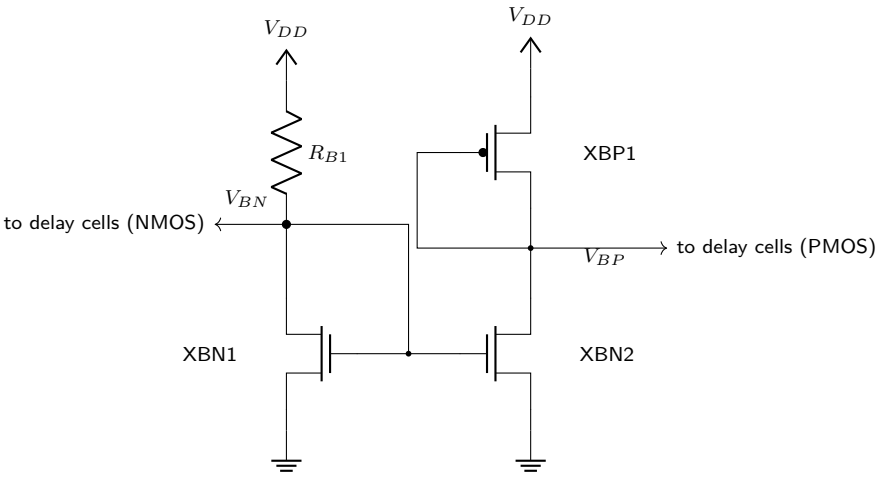


Figure 4: Bias generator, redrawn for readability (the generator is defined in the simulation netlist, not in an Xschem file of the repository): resistor R_{B1} sets the current, XBN1 is an NMOS diode, XBN2 mirrors it into the PMOS diode XBP1, which produces V_{BP}. The starving devices of every delay cell are copies of XBN1 and XBP1.

The generator in Figure 4 replaces the two ideal sources. The resistor R_{B1} from V_{DD} to the NMOS diode XBN1 ($L = 1\,\mu$m, $W = 0.5\,\mu$m) sets

$$I = \frac{V_{DD} - V_{BN}}{R}. \tag{2}$$

The mirror XBN2 copies I into the PMOS diode XBP1 ($L = 1\,\mu$m, $W = 1\,\mu$m), which produces V_{BP}. The starving devices of the delay cells have the same sizes as XBN1 and XBP1, so the cell current should follow I (an expectation, not measured separately). The circuit starts on its own. Calibration at TT, 27 °C, 1.8 V gives $R = 1.247933\,\text{M}\Omega$, for which $V_{BN} = 0.700$ V.

The SKY130 `xhigh_po` resistor model has a sheet resistance of 2000 Ω/sq, $tc_1 = -1.47 \cdot 10^{-3}$ and $tc_2 = 2.7 \cdot 10^{-6}$. In this ngspice setup the model tempco is not applied (the resistor current was identical at −40, 27 and 125 °C), so the main sweep applies the tempco manually to an ideal resistor. A resistor of 1.248 MΩ needs about 850 μm at $W = 1.41\,\mu$m (603 squares); layout area is not evaluated.

3 Verification method

3.1 Screening criteria

Table 2: Screening criteria (my own thresholds, not derived from a settling or timing budget).

#	Criterion	Threshold
A1	Phases are produced in all 45 conditions (5 corners × −40/27/125 °C × 1.62/1.8/1.98 V)	45/45
A2	Overlap: maximum over time of $\min(\phi_1, \phi_2)$	$< 1\,\%$ of V_{DD}
A3	Non-overlap gap (the smaller of the two)	≥ 5 ns
A4	Phase width (the smaller of the two), $T/2 = 125$ ns	≥ 100 ns
A5	Comparator clock rises after ϕ_2 falls and before ϕ_1 rises	45/45
A6	Start-up: width and gap settle within a few cycles; supply ramp 1 and 20 μs	4 corners

3.2 Campaigns

- **PVT.** The deck of the block is patched for corner, temperature and supply (including the clock amplitude); ngspice runs each of the 45 conditions in batch mode. Gap and width are read from `.meas` statements and overlap from the waveform. The same sweep was run for the fixed bias, the replica diodes with an ideal reference current, and the resistor generator with an ideal resistor, with the PDK resistor, and with the manual tempco (main sweep). Two further runs scale R by 0.8 and 1.2 at 12 extreme conditions.
- **Monte Carlo.** One ngspice process per seed, with `.option seed=N` and the mismatch library corners `tt_mm`, `fs_mm` and `sf_mm`, which makes local transistor mismatch random while the global process stays at the corner. 100 seeds at each of three points. The resistor is ideal and has no mismatch.
- **Start-up.** The supply ramps from 0 V in 1 or 20 μs; the clock starts after the ramp, or during it for the 1-μs ramp. Four corners, 12 runs in total.
- **Comparator clock.** The clock for the StrongARM comes from ϕ_{2b} through two inverters, with a 20 fF placeholder load, over the 45 conditions.
- **Closed loop.** Transistor-level loop with a 300 mV sine at OSR 80, 4112 cycles and a 4096-point FFT.

4 Results

4.1 Nominal characterisation

At TT, 27 °C, 1.8 V, with ideal bias $V_{BN} = 0.70$ V and $V_{BP} = 0.71$ V and the real switch load, the gap is 14.27 ns (ϕ_1 falling to ϕ_2 rising) and 14.33 ns (the other direction), and the widths are 110.59 ns and 110.81 ns; skew is below 0.3 ns. Edge times are 0.50 ns rising and 0.28 ns falling for ϕ_1 and ϕ_2, and 0.76 ns and 0.50 ns for the complements. With added wire capacitance of 50/200/500 fF the gap changes only to 14.3/14.5/14.7 ns and the widths to 110.5/110.4/110.1 ns, while the ϕ_1 rise time grows to 0.82/1.77/3.68 ns; overlap stays zero.

4.2 PVT: fixed bias versus the generator

Table 3: PVT sweeps (45 conditions unless noted). Pass means A1–A4 in Table 2.

Sweep	Pass	Gap (ns)	Min width (ns)	Max overlap (% V_{DD})
Fixed ideal bias (35 valid of 45)	17/45	1.94 .. 91.76	32.9	0.031
Replica diodes, ideal reference current	45/45	12.09 .. 16.47	108.4	0.031
Generator, ideal R (no tempco)	45/45	11.21 .. 18.73	106.2	0.031
Generator, PDK resistor (tempco not applied)	45/45	11.34 .. 18.85	106.0	0.031
Generator, R with tc_1/tc_2 (main)	**45/45**	**9.99 .. 20.58**	**104.3**	0.031
Generator, $R \times 0.8$ (12 extremes, no tempco)	12/12	9.14 .. 15.35	109.5	0.012
Generator, $R \times 1.2$ (12 extremes, no tempco)	12/12	13.26 .. 22.06	102.8	0.012

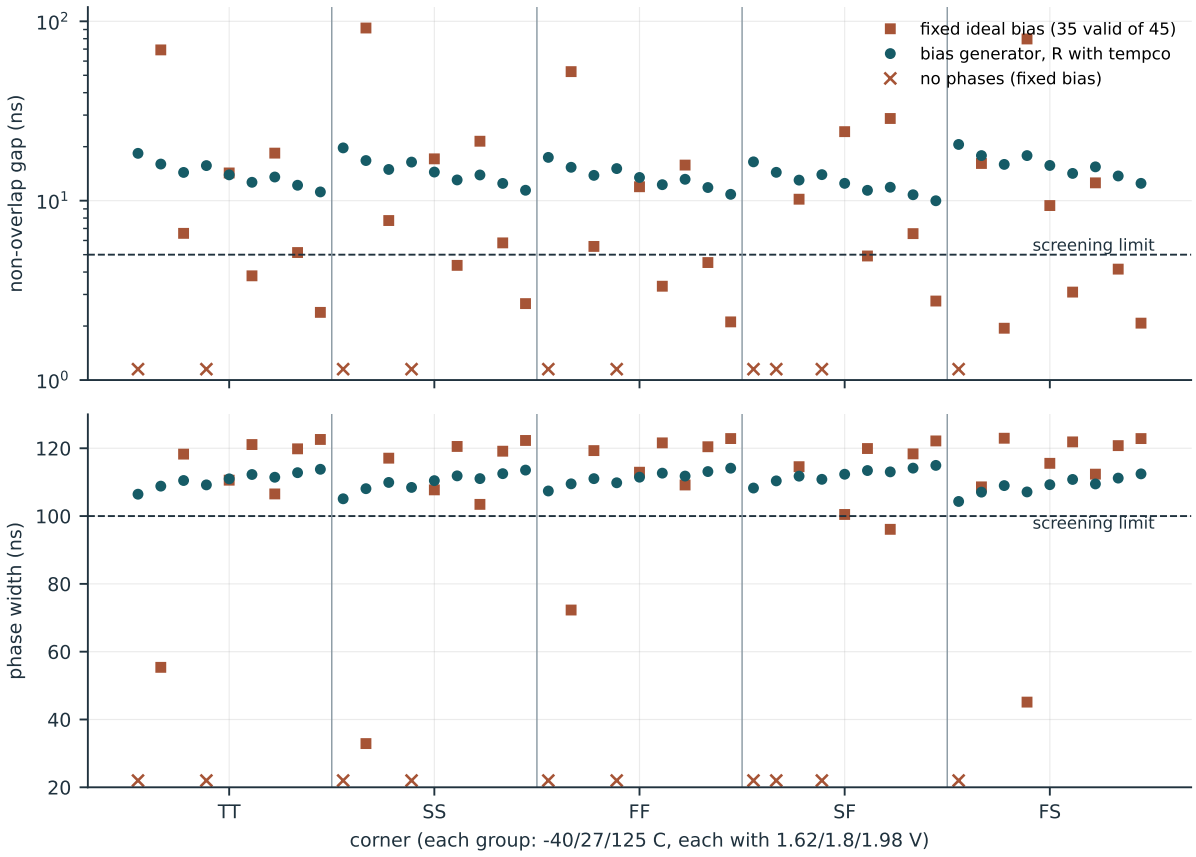


Figure 5: Gap and width in the 45 PVT conditions. Crosses at the bottom mark conditions where the fixed bias produces no phases. The generator keeps every condition on the safe side of both screening limits.

With the fixed bias, 10 conditions produce no phases (nine of them at 1.62 V, plus SF, −40 °C, 1.8 V). Of the 35 valid conditions, 17 pass; 13 fail the gap limit (all at 1.98 V or 125 °C) and 5 fail the width limit. With the generator, the gap spread is a factor 2.06 (from 9.99 ns to 20.58 ns),

against 47 for the fixed bias. The generator with an ideal reference current has a spread of 1.36, so the resistor and its tempco cost some spread. In the main sweep the smallest gap is at SF, 125 °C, 1.98 V; the largest gap and the smallest width (104.31 ns) are at FS, −40 °C, 1.62 V; and the largest overlap is 0.62 mV at FF, 125 °C, 1.98 V. Per-condition tables are in the repository (Appendix B).

4.3 Resistor spread

Scaling R by 0.8 and 1.2 at 12 extreme conditions (without tempco) keeps all runs inside the criteria: gap 9.14–15.35 ns with a smallest width of 109.5 ns for ×0.8, and gap 13.26–22.06 ns with a smallest width of 102.8 ns for ×1.2. The direction matters for the next iteration: a smaller R lowers the gap and widens the phases at the cold, low-supply corner.

4.4 Monte Carlo with local mismatch

Table 4: Local-mismatch Monte Carlo, 100 seeds per point, generator with tempco. Gap and width are means ± standard deviation, with the range in brackets.

Point	Pass	Gap (ns)	Width (ns)
TT, 27 °C, 1.8 V	99/100	14.0 ± 2.85 (7.78 .. 24.69)	109.9 ± 3.20 (94.2 .. 115.1)
SF, 125 °C, 1.98 V (smallest gap)	100/100	10.0 ± 1.56 (6.31 .. 15.53)	114.4 ± 1.69 (106.8 .. 117.4)
FS, −40 °C, 1.62 V (smallest width)	**70/100**	20.9 ± 5.50 (9.95 .. 43.41)	102.0 ± 6.50 (67.1 .. 111.5)

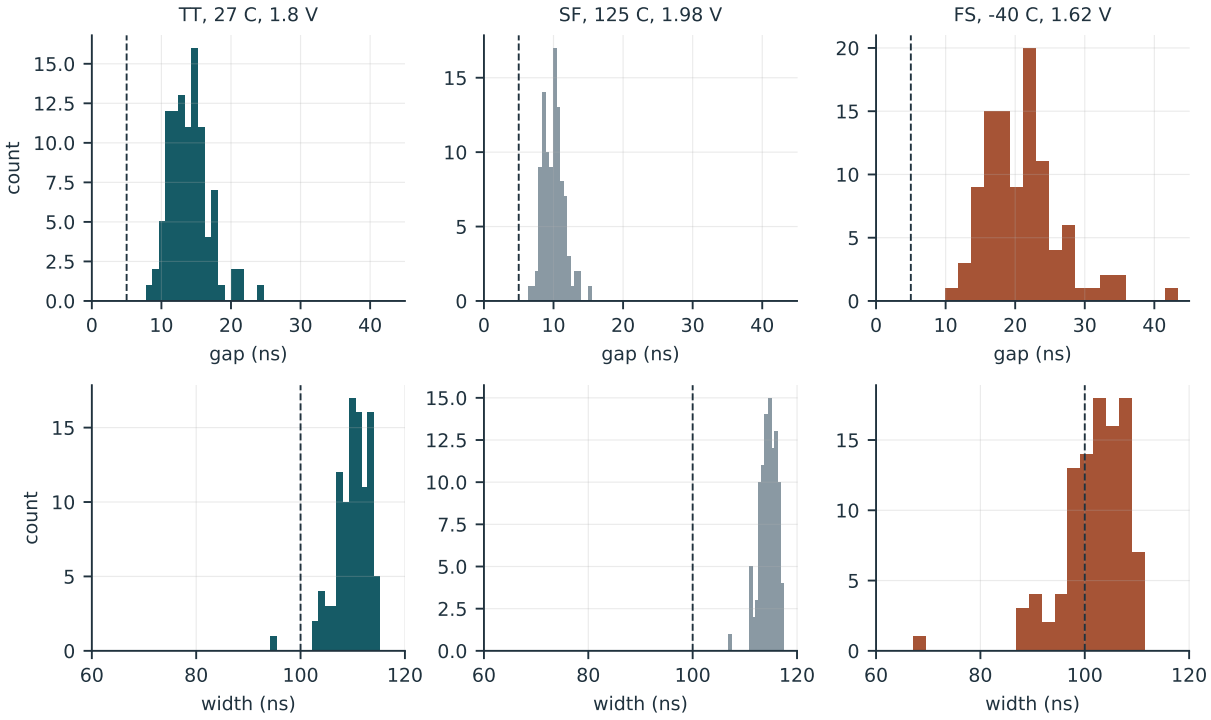


Figure 6: Distribution of the smaller gap (top) and smaller width (bottom) at the three Monte Carlo points. The dashed line is the screening limit.

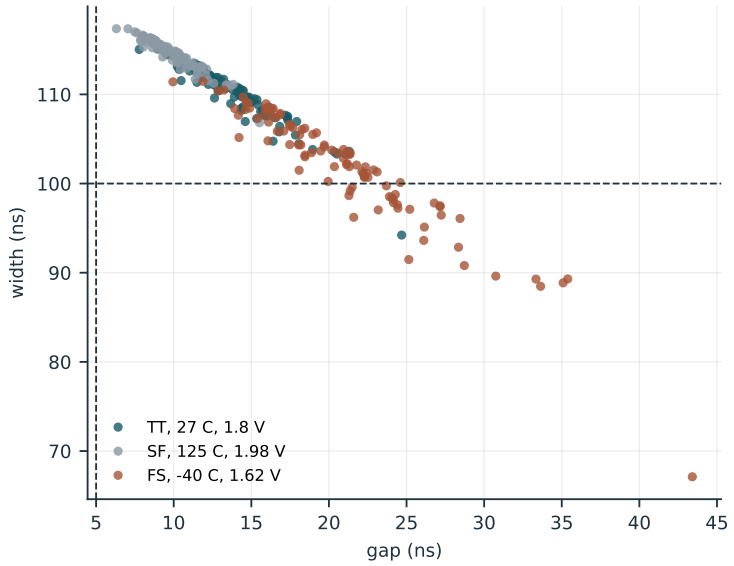


Figure 7: Width against gap for all 300 samples. The samples follow the trade-off of (1); the points in the lower right fail the width limit.

Observations:

- The Monte Carlo means agree with the deterministic values at the same points (10.00 vs. 9.99 ns at SF; 20.9 vs. 20.58 ns at FS). Mismatch scatters the result around the nominal value; it does not shift it.
- All 30 failures at FS, $-40\,^{\circ}\mathrm{C}$, 1.62 V have a width below 100 ns. The gap never falls below 5 ns there. The 95 % interval for a pass rate of 70/100 is about 60–78 %.
- The distributions are skewed (Shapiro–Wilk $p < 0.01$), so "mean -3σ" is not a 3σ claim. The worst measured width of 67.1 ns is far below the Gaussian extrapolation of 82.5 ns.
- At SF the smallest gap is 6.31 ns, only 1.3 ns above the limit.
- The same seeds are the worst in all three runs (seed 2046 gives the smallest width, seed 2038 the smallest gap), and the per-seed gap correlates with a coefficient of 0.99 between the FS and SF runs. This is consistent with a common bias-current component, for example the small mirror devices of the generator, but I did not test it.
- Overlap stays below 0.05 mV in all 300 samples.

The gap and width values are the smaller of the two phases, so a sample can have its smallest gap and its smallest width in different phases. For this reason the sum of the two can fall below $T/2$ (down to 110.5 ns at FS).

4.5 Start-up

Table 5: Start-up from a supply ramp (final values after settling; pulses counted from the clock start).

Corner / T / V_{DD}	Gap (ns)	Width (ns)	Overlap (mV)	Pulses to settle (after / during)
TT / 27 °C / 1.8 V	13.93	110.98	0.004	2 / 3
SS / −40 °C / 1.62 V	19.70	105.11	0.004	2 / 3
FF / 125 °C / 1.98 V	10.85	114.16	0.619	2 / 4
FS / −40 °C / 1.98 V	15.94	108.99	0.004	2 / 3

All 12 runs settle: final width and gap are within 1 ns of their final values, the gap is at least 5 ns, the width at least 100 ns, and the overlap is below 1 % of V_{DD}. The final values equal those of the PVT sweep. With the clock starting after the ramp (ramp 1 and 20 μs), the block is settled from the second pulse; with the clock running during a 1 μs ramp, it takes three or four pulses. The "first pulse" in the first case is not a clock pulse: ϕ_1 follows V_{DD} and stays high during the whole ramp (300 ns to 10 μs wide) until the clock starts. This is a property of the circuit and its effect on the Block 5 switches is not evaluated.

4.6 Comparator clock from ϕ_{2b}

The StrongARM clock is taken from ϕ_{2b} through two inverters with a 20 fF placeholder load. Over the 45 conditions it rises inside the gap in 45 of 45: from 0.28 to 0.73 ns after ϕ_2 falls and 9.69 to 19.68 ns before ϕ_1 rises. The clock is high for 134.6 to 145.0 ns. Block 3 reports a worst-case integrated decision delay of 3.189 ns from its own campaign with 5 ns clock edges; the two numbers use different definitions and were not matched, so this is an indication and not a margin claim.

4.7 Closed loop

Table 6: Closed loop at TT, 27 °C, 1.8 V; 300 mV sine, bin 13, OSR 80, 4112 cycles, 4096-point FFT.

Run	Configuration	SNDR (dB)	ENOB
`sine300_n4096`	ideal phases, ideal comparator clock	75.37	12.23
`f1_sine300_b6_n4096`	Block 6 phases (ideal bias), ideal comparator clock	73.87	11.98
`f5_sine300_n2_n4096`	Block 6 phases (ideal bias), comparator clock from ϕ_{2b}	72.44	11.74
`f6_sine300_b6res_n4096`	Block 6 with the resistor generator, comparator clock from ϕ_{2b}	73.62	11.94

The four rows differ by up to 2.9 dB. The scatter of a single run is about ±1 dB at OSR 80, so I do not claim degradation or improvement between rows. The system target is about 75 dB at OSR 80.

5 Known limits and open items

- **Pre-layout only.** No extraction or parasitics; the clock-node load is a 20 fF placeholder.

- **Criteria.** The screening limits in Table 2 are not derived from the settling budget of Block 5. A pass means a pass against them.
- **Monte Carlo scope.** Local mismatch only, ideal resistor, no global process spread, three points (not combined with the other 42 PVT conditions), 100 seeds each. At FS, −40 °C, 1.62 V the pass rate is 70/100 against the width limit; at SF, 125 °C, 1.98 V the smallest gap is 1.3 ns above the limit.
- **Resistor model.** ngspice does not apply the model tempco or any corner dependence; the tempco was applied manually. Resistor spread is covered only by ×0.8 and ×1.2 at the extreme conditions, without tempco.
- **Closed loop.** Only at the nominal point: one amplitude, one bin, one run per configuration.
- **Supply ramp.** ϕ_1 follows V_{DD} and ϕ_2 can rise to a fraction of V_{DD} (about 0.27 V in an earlier test); the effect on the Block 5 switches is not evaluated. The first cycles after start are not nominal and are assumed to fall inside the 5 μs reset, which is a default value and not a validated requirement. Start-up overlap is measured only after the ramp.
- **Supply current** of the generator is estimated as about $2 \times I$ and not reported separately.
- **Block 3 interface.** The decision delay of Block 3 and the clock timing of this report use different definitions.

6 Decision and next iteration

Date	Decision
2026-11-09	Close Block 6 with option A: accept the 70/100 pass rate at FS, −40 °C, 1.62 V against the width screening limit as a known limit, and defer the next iteration.

Not done, in order of expected effect:

1. Re-centre R to ×0.75–0.8. The deterministic ×0.8 run gives a width of at least 109.5 ns at the 12 extremes (without tempco). A first check at a few extreme conditions, then Monte Carlo at the two corner points, should come before any heavy re-run.
2. Enlarge the generator mirror devices to reduce the spread, if the correlation seen in Section 4 is confirmed to come from them.
3. Derive the width requirement from Block 5 settling instead of using the 100 ns screening value.

A change of R invalidates the PVT, Monte Carlo, start-up, comparator-clock and closed-loop results of this report.

A Reproduction

Environment: ngspice 46, SKY130A models from the PDK in the simulation environment (`libs.tech/combined/sky130.lib.spice`), Python 3 with NumPy and Matplotlib. The tools are in the repository; their `--repo` option points to the simulation tree that contains `blocks/06_Clock_gen/sim/out/netlist/`.

```
python3 tools/pvt_block6.py --repo <sim tree> --out results/pvt_block6_fixed45 --jobs 2
python3 tools/mc_block6.py --repo <sim tree> --n 100 --jobs 2            # TT_mm, 27 C, 1.8 V
python3 tools/mc_block6.py --repo <sim tree> --n 100 --corner fs --temp=-40 --vdd 1.62 --out results/
     mc_block6_fs_m40_1p62
python3 tools/mc_block6.py --repo <sim tree> --n 100 --corner sf --temp 125 --vdd 1.98 --out results/
     mc_block6_sf_125_1p98
python3 tools/startup_block6.py --repo <sim tree> --deck <..._biasgen_res_tc.spice>
python3 tools/startup_report_block6.py --dir <sim tree>/results/startup_block6
python3 top_level/closed_loop/tools/f5_clkcmp_pvt.py --run --jobs 2
python3 tools/summarize_block6.py --repo <repo> --results <sim tree>/results --max-ovl 0.01
```

The generator deck and its calibration come from `tools/bias_gen_block6.py` (see its header for the sub-commands). The closed loop uses `top_level/closed_loop/run_closed_loop_f1.py` with `--b6-bias res`; the exact arguments of each run are stored in its `manifest.json`.

B Evidence index

ID	Content	Repository path
E01	PVT, fixed ideal bias, 45 conditions	`results/pvt/`
E02	PVT, generator sweeps and per-condition tables	`docs/pvt_results_block6.md`
E03	Monte Carlo TT, 100 seeds	`results/mc/tt_27_1p8/`
E04	Monte Carlo SF, 100 seeds	`results/mc/sf_125_1p98/`
E05	Monte Carlo FS, 100 seeds	`results/mc/fs_m40_1p62/`
E06	Start-up, 12 runs	`results/startup/startup_summary.csv`
E07	Comparator clock, 45 conditions	`results/pvt/f5_clkcmp_pvt_summary.csv`
E08	Closed-loop run folders	`top_level/closed_loop/results/closed_loop/`

All paths are relative to `blocks/06_Clock_gen/` except E08. Figures 5–7 were generated from E01–E05.

C Gate-level cells

The cells below are used inside the NONOV (Xschem exports of the files in `blocks/06_Clock_gen/xschem/`). The XOR schematic contains the XNOR gate-level cell `B6_NOR2` as an internal instance.

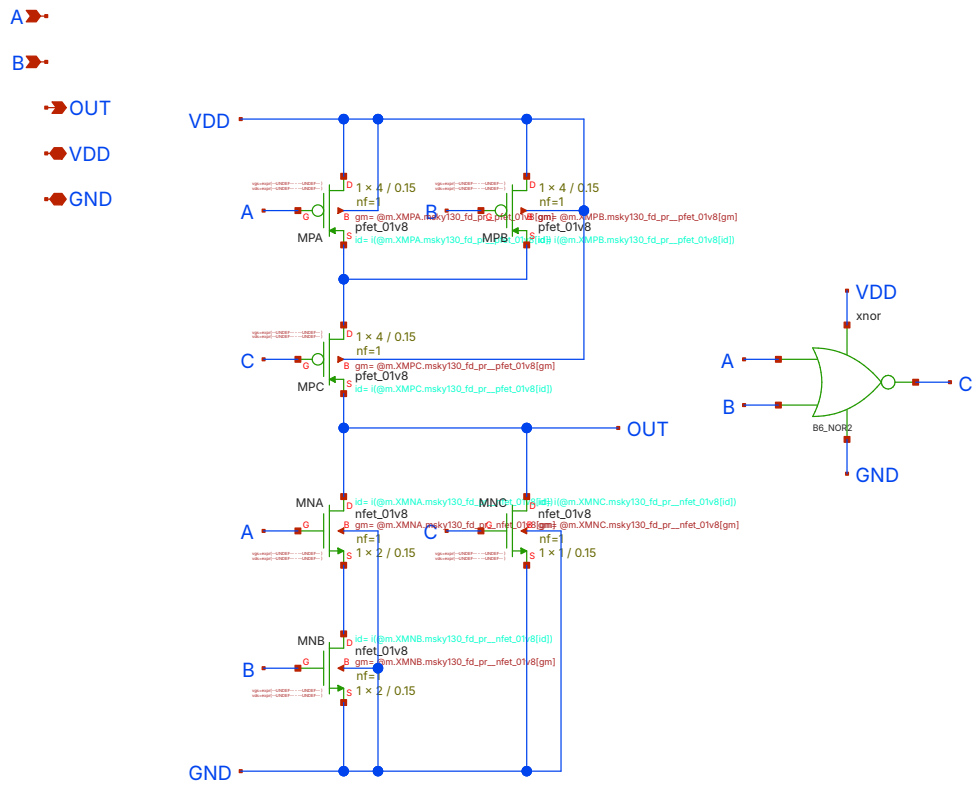


Figure 8: Xschem export of `B6_XOR.sch`.

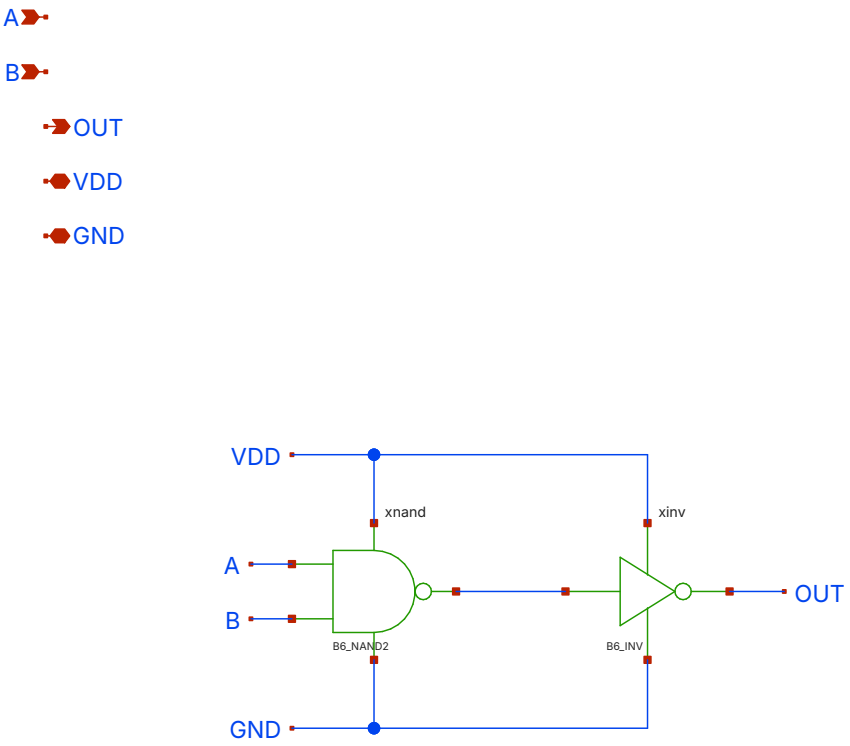


Figure 9: Xschem export of `B6_AND2.sch`.

References

[1] R. Spilka, G. Hilber, A. Rauchenecker, D. Gruber, M. Sams, and T. Ostermann, "Generation of Non-Overlapping Clock Signals without using a Feedback Loop," Johannes Kepler University, Institute for Integrated Circuits, Linz, Austria; IEEE, 2012 (DDECS 2012), ISBN 978-1-4673-1188-5/12.

[2] SkyWater PDK documentation, "Device Details: 1.8 V NMOS FET and 1.8 V PMOS FET." https://skywater-pdk.readthedocs.io/en/main/rules/device-details.html. Online documentation is contextual; the model files actually used are the SKY130A files in the simulation environment (Appendix A).

[3] Ngspice project, *Ngspice User's Manual*. https://ngspice.sourceforge.io/docs.html. The experiment logs identify simulator version 46.